Minimum Viable Video-Text: Flour Sacks, Creative Agent Swarms, and Actuarial Anticombinatorics after Calvino's "Abominable House"

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Abstract

Generative media currently splits inspectable symbolic reasoning from opaque pixel-diffusion video that resists local causal repair, collapses multi-shot continuity, triggers safety refusals on dark literary premises, and saturates human timeline review at swarm scale. Positioning the open-source suite **PAE-04 (*The Burning of the Abominable House*)** alongside its precursor **PAE-03 (*The Flour Tower*)** and companion **Lab 7** films (*By What Light* and the *Bitistics* cycle), this paper advances three contributions grounded in the 1973 convergence of Italo Calvino [12], Clifford Geertz [26], and Stanisław Lem [39]. First, we define **Minimum Viable Video-Text (MVVT / VTX-0.1)**, adapting 1930s studio flour-sack animation exercises [58, 34] and Sybille Krämer's cultural technique of flattening [37, 28] into an addressable representation organized around *chance, memory, and consequence*. Second, drawing an architectural analogy to Anthropic's *Scion* version-control redesign [1], we replace single-viewer 1× timeline scrubbing with atomic state transitions, machine-checkable intent contracts, Explicitly Notated Citations (ENC) [25], and fresh-context blind critics. Third, we execute exact BigInt Markov dynamic programming and two-stage Mackworth AC-3 + CDCL constraint solving (Google OR-Tools CP-SAT) across Calvino's 12^{12} state space. We prove an exact 27 : 25 (1.08) prime-lattice lock with Elena Pierazzo's trillion-variant philology [49], diagnose Calvino's printed 8,874,296,672,256 as an exact 1973 carry slip $12^6(12^6 - 14{,}000)$, quantify a −4.6 pp Šeļa actuarial confound [54] on Widow Roessler's life policy, and prove that closed-world single-actor arson restricts 16 contract states to 11 realizable vectors (102 causal skeletons) until open-world conspirator expansion ($N = 5$) unlocks Skiller's zero-payout equilibrium. Epistemic status is audited strictly throughout, separating MEASURED mathematical proofs from PROPOSED human perceptual and swarm evaluations.

1 1973, a Burned Copybook, and the Exact 27 : 25 Combinatorial Horizon

In February 1973, responding to an exploratory **IBM** inquiry asking whether fiction could be composed with a computer, Italo Calvino published *"L'incendio della casa abominevole"* (*"The Burning of the Abominable House"*) in the Italian edition of *Playboy* [12]. As Esther Calvino's preface to *Numbers in the Dark* documents, computer access proved impractical at the time of composition, so Calvino executed the combinatorial operations manually on paper [15]; only subsequently did he collaborate with Oulipo physicist-mathematician Paul Braffort on formal filtering programs presented at the Centre Pompidou A.R.T.A. session on June 15, 1977 [6] and correspond with former IBM programmer William Skyvington [55]. Keeping the 1973 manual fiction distinct from the 1977 algorithmic prototype clarifies Calvino's foundational premise: four boarding-house inhabitants—Widow Roessler (R), her bald fashion-model daughter Ogiva (O), the disinherited young lord Inigo (I), and the Uzbek wrestler Belindo Kid (B)—perish in an unindexed thirteenth act of arson, leaving behind a charred plastic-shielded copybook whose surviving back index lists twelve alphabetically ordered transitive deeds (*Blackmail, Drugging, Incitement to Suicide, Knifing, Prostitution, Rape, Seduction, Slander, Snooping, Strangling, Threatening with a Gun,* and *Tying and Gagging*) stripped of perpetrators, victims, and page numbers [12]. Hired by insurance agent Skiller, programmer Waldemar maps the $4 \times 3 = 12$ non-reflexive ordered tenant pairs across all twelve categories, opening an unconstrained space of $12^{12} = 8{,}916{,}100{,}448{,}256$ raw assignments prior to chronological ordering.

Notably, **1973** witnessed the simultaneous appearance of four works that frame our contemporary encounter with generative AI: Calvino's *"Abominable House"* [12], his tarot combinatorics in *Il castello dei destini incrociati* [13], Clifford Geertz's *The Interpretation of Cultures*—characterizing cultural analysis as straining to read ensembles of texts *"over the shoulders of those to whom they properly belong"* [26, p. 452]—and Stanisław Lem's *Wielkość urojona* (*Imaginary Magnitude*), whose *"History of Bitic Literature"* inaugurated **bitistics**, the critical study of works authored by nonhuman machines [39]. Half a century later, Digital Humanities faces that exact horizon at scale. Two `MEASURED` arithmetic findings anchor our philological starting point:

1. **The Exact $27:25$ (1.0800) Pierazzo–Calvino Prime Lock:** At DH2025 Lisbon, Elena Pierazzo showed that combinatorial genetic variants across Manzoni's *Il Capitolo dei Frati* generate $8{,}255{,}648{,}563{,}200 = 2^{24} \cdot 3^9 \cdot 5^2$ potential texts [49]. Factoring Calvino's $12^{12} = 2^{24} \cdot 3^{12}$ reveals an exact shared prime spine $\gcd = 2^{24} \cdot 3^9 = 330{,}225{,}942{,}528$ and an exact rational ratio of $27:25 = 1.08$. Yet their methods diverge sharply: where manuscript philology tames proliferation by clustering witnesses into a synthetic *"average manuscript"* centroid [49, p. 1015], detective causality and insurance tail-risk collapse if averaged, demanding lossless state compression paired with quality-diversity archiving [43, 33].

2. **Forensic Solution to the 1973 `8,874,296,672,256` Paratextual Crux:** Printed editions of Tim Parks's translation cite `8,874,296,672,256` solutions [12] (which Wikipedia further truncates to `874.296.672.256`, losing 8×10^{12} [59]). Exact division by $12^6 = 2{,}985{,}984$ resolves this fifty-year discrepancy with zero remainder:

$$8{,}874{,}296{,}672{,}256 \;=\; 2{,}985{,}984 \times 2{,}971{,}984 \;=\; 12^6 \times (12^6 - 14{,}000), \tag{1}$$

   exposing a single thousands-column hand-multiplication carry slip ($85 \to 71$) forced by 8-digit 1972–1973 pocket-calculator overflow above 10^8, erasing exactly 41,803,776,000 "phantom stories" [53].

To make such spaces inspectable across text, motion, and calculation, we introduce **Minimum Viable Video-Text (`MVVT`)**, formalized in the working schema `VTX-0.1` around three temporal primitives: **chance** (configured exogenous disturbances distinct from audience uncertainty or model stochasticity), **memory** (distinguishing persistent world traces, selective character recollections, and production logs), and **consequence** (irreversible physical, epistemic, and contractual state transitions that condition subsequent actions). Extending `[Author]`'s prior LDraw "Linguistic Brick" (`Line Type 1` affine placement) and `LEGOS` spatial-assembly research [38]—which asks whether a visible brick error can be traced to its source line and repaired locally—`MVVT` asks the corresponding temporal question summarized in Table 1.

Table 1: Five inspectability layers required for a Minimum Viable Video-Text (`MVVT` / `VTX-0.1`).

Video-Text Layer	**Inspectable & Locally Revisable Surface**
World & participants	Persistent actor identities, room topology, objects, and initial conditions.
Event account	Ordered or partially ordered deeds $(u \to v, d, \tau)$ with preconditions, chance draws, and effects.
Performance score	Pose, center-of-mass trajectory, timing, squash-and-stretch, contact intervals, and camera framing.
Interpretation ledger	Authored world truth, permitted per-character observations/beliefs, and conditional policy verdicts.
Revision provenance	Source diff, failed gate diagnostics, critic evidence, and deterministic `ENC` state hash [25].

2 Expressive Flattening and the Flour-Sack Abstraction

Why build procedural canvas video-text rather than prompting commercial text-to-video diffusion models? Two empirical and institutional barriers motivate our medium design. First, evaluating

contemporary image and video generators on multi-panel narrative prompts—explicitly including *house fires* and *secretly reading a diary*—Onal, Uren, and Güngör demonstrate persistent breakdowns in character identity, sequential causality, and emotional legibility across cuts, invoking classical animation principles from Johnston and Thomas [45, 34, 4]. Second, Calvino's index enumerates severe interpersonal harms; prompting commercial synthesis APIs to enact human knifing, strangling, or assault rightly triggers content-safety refusals or risks gratuitous depiction.

Following studio practice documented by The Walt Disney Family Museum [58] and codified by Johnston and Thomas [34] (noting this training lineage without claiming Disney invented the exercise), **`PAE-03` (*The Flour Tower*)** and `PAE-04` replace human anatomy with four faceless, volume-conserving, half-filled canvas **flour sacks**: Roessler in rust (`#c95a49`), Inigo in slate (`#6089a8`), Ogiva in ochre (`#c8963e`), and Belindo in sage (`#6f9169`) (kept distinct from *Flower*, a separate plant-performer embodiment transfer test). Stripping away faces and spoken dialogue forces every narrative distinction—hesitation versus commitment, accidental slip versus deliberate shove—into 10–15 geometric parameters governed by area conservation (`INV_VOLUME`: $|\mathrm{Area}_{2D}(i, f) - V_0(i)|/V_0(i) \leq 0.05$) and zero base-skating (`INV_GROUND_CONTACT`).

Crucially, we do not treat flour-sack abstraction as a covert trick to evade provider safety policies, nor claim that burlap sacks preserve the full social gravity of human violence. Instead, we preserve ethical friction: eight deeds stage non-graphic theatrical stagecraft across $8 \times 12 = 96$ procedural rigs in *The Deed Bank* (`player.html`), while the four carnal or compromising deeds retreat behind a **nailed-shut architectural door**, emitting zero interior geometry and registering solely through threshold light slits, state marks, and post-hoc floor traces. As Charlesworth and Warwick argue for destroyed archives at DH2025 [20], this visual absence is load-bearing relational data. Theoretically, the flour sack exemplifies Sybille Krämer's *"cultural technique of flattening"* [37]—operationalized for dramatic networks by Giovannini [28] and pose-silhouette dance tracking by Broadwell, Leonard, and Mimno [8]—turning anatomical reduction into an epistemic instrument that isolates causal kinematics.

3 Creative Agent Swarms after *Scion*: `PEO`, Ologs, and Blind Critics

On October 13, 2026, Anthropic engineering documented rebuilding version control from scratch in *Scion* and its review surface *Antique* after autonomous Claude coding swarms drove monorepo merges from ∼2,500/day to over 30,000/day across 25 million daily checkouts, shattering Git's human-scale assumptions of full local clones, global ref locks, and line-by-line review [1]. Treating Anthropic's infrastructure report strictly as an architectural analogy (and noting no affiliation with `[Author]`'s homonymous `SCION` compiler specification), our **`PAELEO` / `PEO` (*Prompt* · *Ekphrasis* · *Operation*)** framework identifies the parallel biological bottleneck in moving-image tools: traditional non-linear editors assume a solitary human scrubbing a single playhead at 1× clock speed. Once agent swarms generate 54 formal lenses (`PAE-01/02`), 96 sack rigs (`PAE-04`), or trillions of valid cuts, the bottleneck shifts from *generating* to *watching*.

Tracing text–image coupling from *Imagetext* [42, 5] through spatial *Environment Text* to executable *Worldtext*, `PAE-04` coordinates modular scene authoring ('deeds/01..12.js'), momentum-preserving transitions ('trans/00..12.js'), persistent physical marks ('conseq/01..12.js'), and fresh-context critics ('tools/paeleo/reviews/') as mapped in Table 2. Whereas prior iterative video systems such as *VideoAgent* refine pixel plans for robotic manipulation via vision-language feedback [56], `MVVT` binds feedback to an editable symbolic event account—answering Cha's DH2025 keynote call for modular, human-auditable AI workflows [18] and Zweibel's critical-code pedagogy for browser-native JavaScript [10].

To prevent **correlated blind spots**—where the same structured scene record both renders a cut and grades it—an independent critic inspects silent frames before seeing the builder's intent log. Formally, the suite's architecture satisfies a 16-entity Spencer-Brown categorical ontology log (*olog*) across three verified commutative diagrams: (**C1**) brute-force filtering and 28-state `BigInt` dynamic programming commute to the exact integer ('2,663,172,255,744'), whereas centroid averaging fails to commute with insurance liability; (**C2**) room-and-lamp visibility constraints [21] commute with Kablitzian focalization [36]; and (**C3**)

Table 2: Architectural analogy from human-scale Git/NLE pipelines to Anthropic *Scion* and `MVVT` swarms.

Dimension	**Legacy Git / NLE Ritual**	**Anthropic *Scion* + *Antique* [1]**	**`MVVT` Video-Text Swarm (`PAE-04`)**
Checkout unit	Full binary repo / monolithic MP4	Thin daemon cache (200× smaller)	Sparse shot interval + boundary sack states
Mutation atom	Tree commit + global lock	Content-addressed WAL changeset	Declarative transition delta $(u, d, v, [\tau_0, \tau_1])$
Derived views	Local history recomputation	Server-materialized skip lists	Lazily derived X-sheets, volume curves, ledgers
Merge & branch	Serialized pre-merge queue	Optimistic landing + agent repair	Disjoint depth planes merge; conflicts fork *clinamen*
Review gate	Manual 1× line/frame scrub	Intent rules + component "superfans"	Fresh-context blind silhouette critic (sound/labels off)

parameterized Explicitly Notated Citation URIs (`enc://pae04/<commit>/<regime>?reg=watch+read&t0=...#has`
[25, 50] synchronize the theatrical *Watch* register and Tuftean *Read* register onto identical survivor counts.

4 Deadly Serious Numerification: Disambiguated Sieves, the Šeļa Confound, and CP-SAT Contract Proofs

Why does Waldemar feed burned verbs into punch cards at all? As a comparative reading alongside James M. Cain's [9] and Billy Wilder and Raymond Chandler's *Double Indemnity* [60] underscores (without claiming direct influence on Calvino), *"The Burning of the Abominable House"* stages the **actuarial numerification of human life** [44, 23]: four overlapping contracts issued by Skiller—Inigo's house fire policy (P_1, void if Inigo commits arson), Widow Roessler's life policy benefiting Ogiva's child (P_2, void if Ogiva murders Roessler), Ogiva's wig-collection policy (P_3, void if Ogiva torches the wigs), and Roessler's tour-incapacity policy on Belindo (P_4, void if Roessler incites Belindo to suicide)—turn grief and vice into voidability predicates [12].

Because across 'PAE-04' and classroom modules the phrase "the rules" spans distinct constraint layers, Table 3 strictly disambiguates all seven mathematical regimes (`MEASURED` via exact `BigInt` Markov DP and two-stage AC-3 + OR-Tools CP-SAT).

Two sharp theoretical results emerge from Table 3. First, in the 12-slot Markov model (`BELINDO_5RULE_2_66T`), forbidding Belindo from knifing or being strangled never mentions Widow Roessler or Ogiva, yet reduces candidate lethal pairs from 12 to 9, concentrating murder weight onto (Ogiva → Roessler). Extending Šeļa, Haider, and Plecháč's DH2025 proof that formal poetic meters act as statistical confounders [54], our exact DP reveals **double entanglement**: naive local pruning would drop P_2 from $(11/12)^2 = 84.03\%$ to $(8/9)^2 = 79.01\%$, while path-dependent mortality across the 28-state vitality trellis buffers +0.39 pp back to **'79.4

Second, when we require all 12 indexed deeds to instantiate once alongside terminal 13th-act `Arson` in our two-stage Mackworth AC-3 [40] + CDCL CP-SAT engine (sharing the assumption-core architecture of Z3 SMT [22] and ASP stable models [27]), blocking-clause projection onto the lethal-plus-arson skeleton uncovers a strict structural limit:

Proposition 1 (Closed-World 11-of-16 Payout Restriction)**.** *Under closed-world single-perpetrator arson ($N = 4$, $\sum_{u \in V_4} \text{arson}_u = 1$) and post-mortem absorption, exactly* 11 *of the* 16 *payout vectors* $(P_1, P_2, P_3, P_4) \in \{0, 1\}^4$ *are satisfiable across all decoy-noise budgets* $\eta \in \{0, 1, 2, 3\}$*; all four vectors*

Table 3: Disambiguated anticombinatory rule sieves across `BigInt` Markov DP and 13-deed CP-SAT engines (`MEASURED`).

Canonical Regime ID	**DP / Solver Basis**	**Exact Survivors**	**Governing Constraint & Actuarial Discovery**
`RAW_12_POW_12`	1 state (12^{12})	8,916,100,448,256	Unconstrained non-reflexive dyads; P_2 non-void $(11/12)^2 = 84.03\%$, fire 92.8%.
`PARATEXT_8_87T`	$12^6(12^6 - 14{,}000)$	8,874,296,672,256	1973 hand-squaring carry slip erasing 41,803,776,000 phantom histories.
`LEDGER_LEGAL_3_32T`	≤ 28 Markov states	3,316,025,558,016	Marcus/Jokisch stage vitality [41, 35]: dead never act; tied wait 2 slots (37.19%).
`BELINDO_5RULE_2_66T`	≤ 28 Markov states	2,663,172,255,744	Adds Belindo knife/strangle exclusions: **Šeļa confound** drops P_2 to 79.4% (-4.6 pp); fire rises to 95.8%.
`FULL_EPISTEMIC_601B`	Causal memory DP	601,511,297,484	Enforces transitive secret prerequisites (`Snooping` prior to `Blackmail`/`Slander`, 6.75%).
`KABLITZ_LIGHT_3_79Y`	$\leq 85{,}008$ DP states	3.79×10^{24} of 5.46×10^{36}	Room/lamp/trace/witness perception [21, 36] (64%

with $P_1 = P_3 = 0$ (including Skiller's universal-forfeiture target $(0,0,0,0)$) and $(1,0,1,0)$ are formally `UNSAT`.

Proof. Single-actor arson enforces $\text{arson}_{\text{Inigo}} + \text{arson}_{\text{Ogiva}} \leq 1 \implies P_1 \vee P_3 = 1$, ruling out $(0,*,0,*)$. Conversely, $(1,0,1,0)$ requires Ogiva to kill Roessler ($P_2 = 0 \implies \text{arson}_{\text{Roessler}} = 0$) and Roessler to incite Belindo to suicide ($P_4 = 0 \implies \text{arson}_{\text{Belindo}} = 0$) while $\text{arson}_{\text{Inigo}} = \text{arson}_{\text{Ogiva}} = 0$, leaving zero living tenants to ignite the house. □

Table 4: CP-SAT 13-deed skeleton payouts, certified 1-minimal MUS deadlocks, and $N = 4..10$ scaling (`MEASURED`).

13-Deed CP-SAT Regime / Audit	**Skeletons**	**Vecs**	$P(P_1$ Fire$)$	$P(P_2$ Life$)$	$P(P_3$ Wigs$)$	$P(P_4$ Tour$)$
Canonical Story ($\eta = 0$, exact)	102	11/16	0.7059 ($\frac{12}{17}$)	0.6275 ($\frac{32}{51}$)	0.5882 ($\frac{10}{17}$)	0.8235 ($\frac{14}{17}$)
Oulipo Latin-3 Symmetry ($\eta = 0$, exact)	13	6/16	0.8462 ($\frac{11}{13}$)	0.7692 ($\frac{10}{13}$)	0.6154 ($\frac{8}{13}$)	1.0000 ($\frac{13}{13}$)
Lucretian *Clinamen* Staging ($\eta = 0$, exact)	224	11/16	0.7232 ($\frac{81}{112}$)	0.6250 ($\frac{5}{8}$)	0.6339 ($\frac{71}{112}$)	0.8125 ($\frac{13}{16}$)
Open-World Skiller ($N = 5$, single arson)	59	12/16	0.8136	0.8644	0.7966	0.9322
Open-World Collusive ($N = 5, \eta \leq 1$)	60	16/16	0.9667	0.8333	0.5000	0.9500

As summarized in Table 4, CP-SAT also resolves Calvino's open question in *Prose and Anticombinatorics* [6] by proving that strict 4-category subjective traits and degree-3 Latin bi-square symmetry co-satisfy across **exactly** 13 **murder–arson skeletons**, while assumption-core extraction (`SufficientAssumptionsForInfeasibility`) isolates three 1-minimal MUS deadlocks in 107.6–131.2 ms: (*i*) Waldemar's staged-suicide crash (`monotonic_no_staging` ∧ `belindo_strangles_inigo` ∧ `staged_knifing_of_`
whose relaxation via Duncan's Lucretian *clinamen* [24] adds 122 post-mortem staging skeletons ($102 + 122 = 224$); (*ii*) the 3-axiom single-arsonist zero-payout impossibility; and (*iii*) the 2-axiom Oulipo "No Innocents" deadlock. Expanding to $V_5 = V_4 \cup \{\text{Skiller}\}$ (raising the uniform planted-notebook prior from 75.0% to 80.0%) unlocks $(1,0,1,0)$ (12/16 vectors), and permitting collusive multi-actor arson with decoy slack $\eta \leq 1$ unlocks all 16/16 vectors including $(0,0,0,0)$. Across synthetic fraud rings scaling from $N = 4$ ($10^{12.95}$) to $N = 10$ ($90^{12} \approx 10^{23.45}$), AC-3 prunes 1.34–1.36 orders of magnitude (3–27 arc revisions) and CP-SAT finds verified timelines in 117.6–253.2 ms.

5 Tenants with Minds: `LEGOS`, BDI, Constructive Memory, and Focalized Light

To select behaviorally plausible trajectories without surrendering reproducibility to stochastic cloud LLM calls, `PAE-04` synthesizes four cognitive-formal lineages with explicit attribution boundaries:

1. **`LEGOS`, *Growing Entanglements* (`PIE`), and Formalist Dramaturgy:** Reusing `[Author]`'s `LEGOS` grammar (*Location, Entity, Goal, Obstacle, Shift/Solution*) and continuous `PIE` dial balancing practical utility against symbolic obsession, the agent layer operationalizes Russian Formalist Boris Yarkho's self-interest saturation metrics [28, 35] alongside Solomon Marcus's binary spectator-stage matrices [41].

2. **Bratman / Rao–Georgeff BDI and Constructive Scene Simulation:** Explicit **Belief–Desire–Intention (BDI)** commitments are anchored in Michael Bratman's philosophy of practical reason [7] and Rao and Georgeff's computational architecture [51]. Independently—without attributing BDI to Hassabis—we draw on Demis Hassabis and colleagues' cognitive-neuroscience findings on constructive episodic simulation [29, 31] and 4-protagonist personality modeling across 12 scenarios [30], paired with Bayesian inverse planning [3, 19] and DeepMind's *Concordia* Game-Master pattern [57]. Across 333,333-run MCTS batches, agent believability climbs from 0.29 (random) to 0.35 (ledger-filtered) to 0.52 (memory-and-motive guided), surfacing *The Most Likely Story* (`film 1,541,585,241,617`).

3. **Memory Streams (*Generative Agents* vs. `ai-town`) and GOLEM Beliefs:** Adapting Joon Sung Park and colleagues' Smallville loop of timestamped memory streams, reflection, and planning [46]—cited distinctly from the open-source `a16z-infra/ai-town` starter kit [2]—browser modules (`memory/{R,I,O,B}.js`) update grievance and suspicion as typed `crm:E13_Attribute_Assignment` propositions compatible with the GOLEM ontology [47, 16].

4. **Kablitzian Focalization (*By What Light*) and Genettean `T ZERO`:** Whereas Pichler and Pagel show at DH2025 that zero/few-shot and DSPy-prompted LLMs stall at $F_1 = 0.25$–0.76 when detecting literary focalization [48], *By What Light* encodes Andreas Kablitz's split between *histoire*-internal room/lamp perception and external dramatic irony [36] over Alison Knowles and James Tenney's four-slot *House of Dust* grammar [21]. Pressing `T ZERO` in `film.html` enacts a Genettean descriptive pause (`fabula` velocity = 0) [61], freezing action to expose all 12 counterfactual branches across 1,728 openings.

6 Reading Over the Machine's Shoulder: Geertz, MAP-Elites, and Lem's *Semascope*

Adapting Geertz's reading metaphor [26, p. 452] as a methodological analogy—observing public state transitions, revisions, and counterfactuals rather than claiming transparent access to hidden neural weights or equating four scripts with a human culture—`PAE-04` pairs a theatrical **Watch** register (Lotte Reiniger silhouette staging, `#a3321f` brick-red marks) with an analytical **Read** register (Tuftean slopegraphs and sluice-gate bars) [17, 52, 32]. Rather than collapsing survivors into one optimal plot, `atlas.js` illuminates 6,030 diverse trajectories via **MAP-Elites** [43] along Calvino's *Six Memos for the Next Millennium* [14] (*Lightness, Quickness, Exactitude, Visibility, Multiplicity, Consistency*), realizing Jacomy and colleagues' DH2025 call for *irreductionist* visual exploration [33].

This public observability places `PAE-04` in direct dialogue with Stanisław Lem's *"History of Bitic Literature"* [39], where human prompt-givers act as "indirect authors" across *Monoetics, Mimesis, Sophocrisis,* and transhuman *Apostasy*. In Lem's verified text (Section 3.B), Ellias and Semmelberg's electron **`semascope`** projects a corpus's semantic body as woven *"burning threads"* forming a geometric **`semature`** held in equilibrium (**`semastase`**)—exposing an unclosed topological gap in Dostoevsky filled by *The Girl*, while diagnosing recursive overload in failed continuations of Kafka's *The Castle* [39]. Keeping Lem's `semascope`/`semature` terminology philologically distinct from our own methodological extension (born from a productive voice-transcription slip between *semascope* and *semaphore*), we pair the **semascope** manifold with a dual **`Burn_Semaphore`**: graphically, two top flour-sack corners signal alarm or complicity like optical semaphore arms; structurally, a concurrency gate trips the Thirteenth Act when actuarial contradictions or echo-loops collapse `semastase`.

7 Controlled Falsification Protocol, Epistemic Ledger, and Open Seams

Controlled First Experiment (`PROPOSED / UNMEASURED`): Before claiming perceptual or swarm superiority over full-house runs, `VTX-0.1` specifies a falsifiable single-root A/B test: two 12–20 s one-room scenes with two faceless flour sacks, one lamp, a shared camera, and the same terminal fire. **Comparison A (Goal/Intention)** holds memories and exogenous chance draws fixed while varying one authored goal—reaching to assist the other sack versus deliberately tipping the lamp. **Comparison B (Focalized Memory)** holds the physical tipping fixed while varying whether the second sack witnesses the act in a lit room or enters afterward. Naive viewers judge silent, uncaptioned playback against four synthetic demonstration clauses (P_1: confirmed fire; P_2: accidental tip; P_3: deliberate tip; P_4: corroborated witness evidence, outputting `PAY | NO_PAY | UNKNOWN`), while single-pass, single-agent-with-critique, and multi-agent swarm builds are compared under matched compute budgets.

Table 5: Epistemic audit ledger separating verified computations from open empirical and archival gates.

Claim or Component	**Status**	**Verification Basis or Required Next Gate**
$27:25$ Pierazzo–Calvino prime lock & $12^6(12^6 - 14{,}000)$ slip	`MEASURED`	Exact Python/Node `BigInt` factorization and zero-remainder division.
28-state Markov DP (3.32T $\rightarrow$ 2.66T) & -4.6 pp Šeļa shift	`MEASURED`	Exact transition-trellis marginalization (84.03% $\rightarrow$ 79.4%).
CP-SAT 102/13/224 skeletons, 11/16 proof & 3 MUS cores	`MEASURED`	Exhaustive OR-Tools CP-SAT blocking clauses & deletion-minimal MUS.
Olog commutation (`C1--C3`) & `enc://pae04/...` URI hashing	`STRUCTURAL`	Endpoint equivalence check & deterministic FNV-1a state hash.
Silent flour-sack intent legibility ($\geq 80\%$) & swarm speedup	`PROPOSED`	Requires blind human viewer trials & matched-budget A/B swarm runs.
Origin of 14,000 slip (1973 *Playboy* vs. 1993 Mondadori)	`UNKNOWN`	Requires physical collation against Feb. 1973 Italian *Playboy* pp. 58–63.

True to Oulipian workshop practice, four living seams and three divergent research branches remain open across *The Whole House* (Table 5): (**1**) appending live Skiller/Waldemar policy verdict cards to `film.html`; (**2**) unlocking `ENC`-hashed cross-repository links between classroom `2700-fa26` labs and

`bolas` case pages; (**3**) testing embodiment transfer from canvas flour sacks to *Flower*; and (**4**) exploring three orthogonal extensions—*Branch* α (`phantom-14000.html`, staging the 41,803,776,000 carry-slip stories as Skiller's redacted punch-card drawer), *Branch* β (*Adversarial Underwriting*, a two-player game weaponizing non-local Šeļa constraint entanglement), and *Branch* γ (*Shut-Door Acoustic Sortes*, reconstructing `crm:E13` beliefs solely from muffled hallway rhythms and floor traces at `T ZERO`).

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